

Standards-Based Digital Learning Credentials: Strategic Report for University of Basel

Bottom line: University of Basel should run a bounded pilot now, using Implement Consulting Group as the delivery and endorsement partner, but Basel should remain the single formal issuer. The pilot should issue **Open Badges 3.0 credentials conforming to W3C Verifiable Credentials 2.0**, alongside a human-readable certificate, with university-controlled issuer identity, signing keys, revocation/status, downloadable credential files, public verification and wallet export.

1. Executive summary and one-page recommendation

Decision

Question	Recommendation
Pilot now?	Yes—but as a controlled infrastructure and learner-experience pilot, not an institution-wide credential transformation.
Use case	One professional-learning cohort, such as Change Management Practitioner .
Partner	Implement Consulting Group , because the pilot explicitly tests university–external-provider credential governance.
Formal issuer	University of Basel.
Implement’s role	Delivery partner, co-designer and, ideally, issuer of a separate OB3 EndorsementCredential.
Standard	Open Badges 3.0 over W3C VC 2.0.
What learners receive	A visual certificate/badge, public verification page, downloadable signed JSON credential, LinkedIn link and wallet-export option.
Technology	A standards-compliant managed platform or tightly bounded issuer service—not a new university-wide custom platform.
First-pilot exclusions	No ECTS, admission rights, degree credit, automatic HR decisions, blockchain, selective disclosure, Swiss e-ID dependency or multi-university recognition claims.
Go/no-go after pilot	Continue only if credentials pass independent verification, users understand them, sharing exceeds PDF-only behaviour, administration is manageable and governance is accepted.

Why now

The technology is no longer purely speculative. **W3C Verifiable Credentials 2.0 became a W3C Recommendation in May 2025**, and OB3 now defines signed learning credentials using that model. OB3 represents a single achievement; CLR can bundle achievements into longitudinal records; third parties can add endorsements. Operational implementations now

include Open Badge Factory's OB3 production service, Credly ingestion of OB2/OB3, Accredible OB3/VC export and ingestion, and DCC reference wallets and verifiers. [\[w3.org\]](https://w3.org), [\[1edtech.org\]](https://1edtech.org) [\[openbadgefactory.com\]](https://openbadgefactory.com), [\[learn.credly.com\]](https://learn.credly.com), [\[accredible.com\]](https://accredible.com), [\[github.com\]](https://github.com)

Switzerland is moving, but unevenly. ETH Zürich operates Open Badges 2.0 for continuing education; ZHAW institutionalised a PDF-plus-badge microcredential model in September 2025; BFH demonstrated a continuing-education VC in the Swiyu wallet; and swissuniversities has published a common—but explicitly non-final—microcredential understanding. This is sufficient evidence to justify learning by doing, but **not evidence of a mature national portable-learning-record network**.

Strategic conclusion

Digital credentials are currently **three things at once**:

1. **Operationally**: a better, verifiable and more shareable certificate channel.
2. **Emergently**: a machine-readable exchange layer between learning providers, learners, wallets and verifiers.
3. **Not yet**: a broadly trusted, routinely consumed European employment and higher-education infrastructure.

Basel should therefore **pilot the infrastructure while making no inflated recognition claims**.

2. Standards and terminology

Instrument	What it actually is	Verification/portability
PDF/electronic certificate	Human-readable document. A normal PDF has no standard machine-readable achievement model.	File possession; authenticity may depend on visual features or separate verification.
URL/QR verification	A pointer to an issuer/vendor-hosted record. QR is transport, not a standard.	Convenient, but usually depends on the host remaining available. Basel currently uses online verification for academic documents. [unibas.ch]
Proprietary digital badge	Vendor-defined page, image and metadata.	May be shareable and verifiable, but portability depends on export formats and vendor policy.
Open Badges 2.0	JSON-LD achievement assertion, commonly hosted or baked into PNG/SVG.	Portable across OB2-capable systems, but not natively a W3C VC. ETH explicitly uses OB2.
Open Badges 3.0	1EdTech learning-achievement profile built on W3C VC 2.0 ; includes achievement, criteria,	Cryptographically signed, independently verifiable and wallet-compatible when implementations agree on proof/status methods. [1edtech.org] , [github.com]

	earner, issuer, alignment and evidence.	
W3C VC 2.0	General issuer–holder–verifier data model, not an education credential definition.	Supports embedded Data Integrity proofs or enveloping JOSE/COSE proofs, status, schemas and presentations. [w3.org]
Microcredential	Policy/academic concept describing a small assessed learning unit; not a file format.	Can be represented as PDF, OB3, EDC or another format.
EDC/EDCI	European Digital Credentials for Learning and the associated Commission infrastructure. EDC is an ELM application profile.	JSON-LD VC plus mandatory electronic seal, issued/viewed/stored through EU or self-hosted tooling. [europass.europa.eu] , [europass.europa.eu]
Europass	EU user-facing portfolio, wallet and sharing environment; not itself the credential data standard.	Stores EDCs and supports wallet deposit, share links and PDF export. [europass.europa.eu] , [europass.europa.eu]
Wallet/backpack	Holder-controlled or account-based software for storing and sharing credentials.	A wallet does not itself guarantee interoperability or issuer trust.
Blockchain credential	Credential or hash anchored to a distributed ledger.	Blockchain is optional infrastructure, not required by OB3 or W3C VC ; W3C permits web, database, DID and ledger registries. [w3.org]

OB3 signing, identity, status and dual issuers

An OB3 credential contains one issuer value and is signed using a verification method controlled by that issuer. Verification establishes **authorship, integrity, conformance and current status—not whether the educational claim is substantively true**. A verifier must still decide whether it trusts that issuer for that achievement. [\[w3.org\]](https://w3.org)

credentialStatus is optional; the W3C model supports revocation and suspension and gives Bitstring Status List as an example while warning against per-person “phone-home” status designs that reveal verification activity. [\[w3.org\]](https://w3.org)

OB3 does not provide an arbitrary array of co-issuers. Both W3C VC and OB3 model a single formal issuer; multiple proofs do not automatically create multiple issuing authorities. The standards-native collaboration pattern is:

- one formal issuer;
- partner named in achievement metadata/branding; and
- a separately signed EndorsementCredential.

1EdTech explicitly supports third-party endorsements of an issuer, achievement programme or individual credential. [\[1edtech.org\]](https://1edtech.org)

3. Switzerland landscape

Institution/infrastructure	Verified state by Sep 2026	Standard/technology	Assessment
ETH Zürich	Badges for successfully completed short continuing-education courses and microcredentials; LinkedIn sharing, unique verification URL and wallet storage.	Open Badges 2.0 , ETH-hosted; Open Badge Passport mentioned.	Operational, but legacy standard.
ZHAW	Institutionalised Sep 2025; intended coverage of individually taken 1–9 ECTS continuing-education courses by Sep 2026. Automated PDF plus badge, ZHAW-hosted persistent link and LinkedIn workflow. [blog.zhaw.ch]	Open Badge version not stated publicly.	Strongest Swiss institutionalisation evidence, but exact technical conformance is unclear.
BFH	2025 proof of concept for “Data Management & Open Government Data” continuing-education certificate in Swiyu. [bfh.ch]	Verifiable Credential on Swiss trust infrastructure.	Technically relevant; pilot, not institutional rollout.

University of Zurich	OpenOlat documentation supports course, component and global Open Badges; manual/automatic awarding, export/import and LinkedIn display. [docs.olat.uzh.ch]	Open Badges; version not evidenced.	Platform capability ≠ institution-wide programme.
University of Basel	Academic records are downloadable and can be authenticated through verify.unibas.ch ; public evidence reviewed did not show OB3/EDC issuance. [unibas.ch]	PDF plus verification service.	Strong baseline for comparison, not portable VC infrastructure.
HSLU	Deep-tech microcredentials are a pilot, non-credit, with completion certificates after assessment. [hslu.ch]	No public standard evidenced.	Microcredential naming should not be mistaken for OB3/VC.
EPFL/SWITCH/Educa	Digital maturity-certificate work and Swiss wallet/identity integration have been publicly described as proof-of-concept infrastructure. [switch.ch]	Swiss e-ID/Swiyu trajectory.	Nationally important, but not evidence of institutional OB3 operation.
Bern, St. Gallen, FHNW, SUPSI and several other UAS	Public searches found programmes, certificates and microcredential discussion, but no	Unverified.	Do not treat “digital certificate” as standards evidence.

	sufficiently specific evidence to classify an institutional OB3/EDC implementation.		
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Swiss policy and infrastructure

swissuniversities' working paper seeks coherence and credibility for Swiss higher-education microcredentials but explicitly says its position is **not final**. Switzerland therefore has emerging principles, not one binding national credential architecture or ecosystem owner.

[\[swissuniversities.ch\]](https://www.swissuniversities.ch)

The Swiss e-ID vote of 28 September 2025 created a stronger foundation for digital identity. SWITCH stated that it plans to connect the government e-ID with lifelong SWITCH edu-ID and, with Educa, support educational records in Swiyu while pursuing European compatibility. That is an important future integration path, but a Basel pilot need not wait for it. [\[switch.ch\]](https://www.switch.ch)

Legal design requirements: perform a Swiss FADP/privacy assessment; minimise embedded personal data; separate the public share page from the downloadable credential; define retention and deletion rules; and avoid public birth dates. W3C specifically warns about correlatable subject IDs, PII in credentials, wallet data mining and verification mechanisms that expose holder activity. [\[w3.org\]](https://www.w3.org)

4. Europe / EU landscape

EDC offers free Commission services and open-source issuer, viewer and wallet components. Credentials can cover formal education, training, volunteering and non-formal learning; the electronic seal is mandatory, and the Europass viewer checks authenticity and integrity. EDC 3.0 went live in December 2025, adding privacy terms, accessibility improvements and updated APIs; its open-source release was scheduled for 2026.

[\[europass.europa.eu\]](https://europass.europa.eu), [\[europass.europa.eu\]](https://europass.europa.eu) [\[futurium.europa.eu\]](https://futurium.europa.eu)

Important interoperability caveat: current Europass developer documentation still describes EDC as aligned to **W3C VC 1.1**, while newer ELM material describes a move toward VC 2.0. Therefore “both are VCs” does **not** mean an OB3 wallet will automatically ingest an EDC or vice versa. ELM and OB3 use different application profiles, vocabularies, validation rules and trust mechanisms. [\[europass.europa.eu\]](https://europass.europa.eu)

Dimension	OB3	EDC/Europass	Proprietary platform credential
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Core purpose	One learning achievement	European learning qualification/credential	Vendor-defined
Data model	OB3 + VC 2.0	ELM EDC profile; documentation spans VC 1.1/2.0 transition	Varies
Trust anchor	Issuer identifier, key and verifier policy	Electronic seal and European trust framework	Vendor/issuer account
Status	Optional VC status mechanism	EDC viewer/infrastructure checks	Vendor-specific
Learner storage	Any compatible wallet/file	Europass wallet or compatible implementation	Vendor profile/wallet
Social sharing	Usually via rendered URL	Share links/PDF export	Usually polished
Strength	Open, achievement-centric, supplier-neutral	Rich European semantics and institutional trust	Fastest UX and administration
Weakness	Fragmented wallets/trust registries	Institution-heavy, seal and ELM complexity	Lock-in and uncertain export fidelity

EBSI remains relevant as European trust/infrastructure experimentation, but **EDC, ELM, eIDAS seals and the forthcoming EUDI-wallet ecosystem—not blockchain—are the more decision-relevant Basel alignment points.**

5. European higher-education examples

Example	Credential/use case	Maturity
University of Oxford	Accredible pilot for internal professional learning, including Confident Manager and Project Management Essentials; wallet, LinkedIn/CV sharing and long-term access after leaving Oxford. [ox.ac.uk]	Operational pilot; highly comparable to Basel's proposed use case.
ECIU University / University of Twente	E-sealed microcredentials issued through EDC for challenge-based learning across a European University alliance. [universite...derland.nl]	Strong EDC reference implementation.
SURF edubadges, Netherlands	National higher-education badge platform linked to lifelong eduID, with backpack, consent and external authenticity checking. [surf.nl]	Most convincing sector-level infrastructure example.
Imperial Executive Education	Verified digital completion certificates for executive programmes. [execed-onl...rial.ac.uk]	Operational certificate channel; exact standard/vendor not stated.

MIT Learn / DCC	Signed OB3 samples, QR links, VerifierPlus and Learner Credential Wallet interoperability. [github.com]	Important reference architecture, not European production evidence.
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Public evidence reviewed for Cambridge, UCL and several German, Nordic, Austrian and French institutions was often limited to proprietary badges, isolated courses or projects without exact standards. These should not be presented as equivalent to SURF or EDC issuance.

6. Professional training and consulting

Provider	Verified practice	Portability assessment
Implement Consulting Group	No public standards-level badge/VC programme was identified in the reviewed material.	Opportunity , but absence of evidence is not proof that no internal capability exists.
Deloitte	Credly badges with skills, training, assessment and engagement criteria; badges explicitly carry limitations regarding reliance. [credly.com]	Shareable/vendor-network credential. Exact per-badge OB version not shown.
PwC	Credly learning badges can encode hours, application and assessment criteria. [credly.com]	Strong capability-evidence design benchmark.
Microsoft	Credly-hosted secure and verifiable badges for certifications and exam-based achievements. [credly.com]	Mature professional-brand ecosystem.
AWS	Credly badges, public links, social sharing and active/expired certification status tied to recertification. [aws.amazon.com]	Excellent verification and lifecycle benchmark.
McKinsey Academy, BCG/BCG X, EY, KPMG, Accenture, Capgemini, BearingPoint and IMD	Public searches found learning programmes or certificates, but insufficient consistent evidence to classify exact standards, wallet support and portability.	Treat as benchmarking leads, not verified OB3 cases.

The lesson for Implement is not “consultancies already use OB3 everywhere.” It is that major providers use credentials to make **assessment criteria, skills, validity and brand visible**. A Basel–Implement credential adds value beyond marketing only if it remains downloadable, independently verifiable and governed by transparent achievement criteria.

7. Vendor and open-source landscape

Option	Evidence and fit	Principal concern
Accredible	OB3 and W3C VC export/ingestion, wallet, preserved signatures, white-label experience; Oxford uses it for a professional-learning pilot. [accredible.com] , [ox.ac.uk]	Confirm issuer-key control, EU/Swiss hosting and full export contractually.
Credly/Pearson	Strong professional network, OB2/OB3 ingestion and workforce API; used by AWS, Microsoft, Deloitte and PwC. [learn.credly.com] , [aws.amazon.com]	Network-centric model may create dependence.
Open Badge Factory/Passport	OB3 in production since Sep 2025 and 1EdTech-certified; European wallet ecosystem. [openbadgefactory.com]	Provider itself acknowledges unresolved wallet/policy questions.
Canvas Credentials/Badgr	LMS-integrated badging and OB3 direction.	Verify actual certified profile/version in procurement.
Digitary/Parchment	Strong academic-document exchange heritage.	Public evidence reviewed was insufficient for exact OB3/EDC classification.
Sertifier/Certopus/Certifier	Accessible certificate/badge SaaS.	Treat “Open Badges support” as unverified until version, export and conformance are demonstrated.
Europass/EDC tooling	Free hosted and open-source issuer, viewer and wallet stack. [europass.europa.eu] , [europass.europa.eu]	Electronic-seal, ELM modelling and Swiss participation complexity.
DCC stack	Open-source issuing dashboard, LCW and VerifierPlus; active repositories. [github.com] , [github.com]	Reference components require integration, operations and support ownership.
Certo	AGPL-3.0 self-hosted OB3 platform with issuance, sharing and revocation. Its own documentation identifies missing proof-suite	Suitable for lab work, not yet Basel production infrastructure.

	breadth, external status-list resolution, SSO and multi-tenancy. [github.com]	
CredTrail	Emerging open-source OB3 platform.	Very new; institutional operating evidence remains weak.

Prototype assessment: your described architecture—OB3, W3C VC, issuer identity, cryptographic proof and DCC/MIT tooling—is directionally aligned with the current standards stack. The most important production gaps to test are not badge rendering but **key custody, issuer trust, status, durable hosting, subject identity, standards conformance and wallet export**. I could not retrieve the public repository contents through the available GitHub retrieval path, so I have deliberately not asserted implementation details beyond your description.

8. Adoption barriers

- **Technical:** incompatible proof suites, VC 1.1/2.0 transition, wallet fragmentation, key rotation, status availability, durable JSON-LD contexts and identity binding.
 - **Organisational:** who may define an achievement, approve issuance, operate keys, correct errors and revoke credentials.
 - **Legal/policy:** lawful participant-data processing, processor agreements, public disclosure, retention, accessibility and misleading recognition/credit claims.
 - **UX:** learners understand LinkedIn links and PDFs; wallets and raw JSON remain unfamiliar. VerifierPlus itself explains why human-readable public pages remain necessary even when the credential is independently portable. [\[github.com\]](https://github.com)
 - **Market recognition:** cryptographic validity proves issuer authorship, not employer acceptance. Trust remains a policy and reputation decision. [\[w3.org\]](https://w3.org)
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9. University of Basel strategic options

Option	Cost/effort	Value	Recommendation
Continue PDF + verify.unibas.ch	Low	Familiar and credible, but limited machine portability	Retain as baseline/control.
Proprietary badge platform	Low–medium	Best immediate learner UX and analytics	Good pilot route only with OB3 export and key/identity safeguards.

Lightweight self-hosted issuer	Medium–high	Maximum control and learning	Use prototype/reference stack for testing, not first production.
Full EDC/European alignment	High	Strong European semantics and institutional trust	Monitor/test separately; premature as sole pilot route.
Wait for Swiss national infrastructure	Low now, opportunity cost high	Potential future integration	Do not wait; design migration hooks.

10. Basel + Implement pilot recommendation

Governance model

- **A—Basel issuer, Implement delivery partner:** clean and low complexity.
- **B—Implement issuer, Basel recogniser:** weakens Basel’s control and may imply less formal university responsibility.
- **C—one Basel issuer, jointly defined/co-branded achievement, Implement endorsement: recommended.**

The credential should say that the programme was delivered with Implement, while the cryptographic issuer is Basel. If Implement is ready to manage its own organisational signing identity, it can issue a separate EndorsementCredential; otherwise begin with signed Basel issuance plus clearly governed co-branding. 1edtech.org

Minimum credible architecture

1. Basel-controlled did:web or HTTPS issuer identifier on a university domain.
2. Signing key in managed KMS/HSM, with named key owner, rotation and incident procedure.
3. OB3/VC2 credential containing achievement name, criteria, learning outcomes, assessment, workload, evidence policy, issue date and participant name.
4. Privacy-preserving status list for revocation/correction.
5. Email claim link; human-readable verification page; downloadable signed credential and accessible PDF rendering.
6. One-click LinkedIn addition plus tested export to at least two independent tools—e.g. VerifierPlus and LCW/Open Badge Passport.
7. A one-cohort **20–40 learner** pilot—directional recommendation, not an evidence-derived threshold.

Success measures

- 100% pass an independent OB3 verifier;
- claim, download, LinkedIn-sharing and wallet-export rates;
- learner understanding of “verified” versus “recognised”;

- employer/recruiter comprehension interviews;
 - issuance/correction administrative time;
 - zero unresolved identity mismatches or key/status incidents;
 - evidence that the portable file remains verifiable outside the issuing platform.
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11. Business case

Benefit	Evidence strength	Assessment
Easier LinkedIn/CV visibility	Evidenced	ETH, ZHAW, Oxford and AWS explicitly support it.
Faster authenticity checks	Evidenced	OB3/VC signatures and public verification are operational. [1edtech.org] , [github.com]
Reduced certificate administration	Plausible	Automation is available, but Basel-specific savings must be measured.
Basel/Implement brand exposure	Evidenced but secondary	ZHAW reports strong social sharing; vendor evidence supports branded share pages. [blog.zhaw.ch] , [accredible.com]
Richer evidence of learning	Evidenced technically	OB3 supports criteria, evidence, alignments and skills. [1edtech.org] , [github.com]
Portable lifelong record	Plausible/emerging	Wallets and national examples exist, but cross-ecosystem consumption remains fragmented. [surf.nl] , [lcw.app]
Employer hiring automation	Weak/speculative for this pilot	No evidence that Swiss employers routinely ingest OB3/EDC into HR systems.
Formal academic recognition	Weak unless governed separately	A digital format does not create credit, admissions or qualification status.

12. Risks and unresolved questions

1. Can Basel legally and operationally control the signing identity and keys?
2. Which proof/status combination passes 1EdTech, DCC and selected vendor verifiers?
3. Will Implement operate an endorsement identity, or only provide co-branding?
4. Can credentials be exported with original signatures after contract termination?
5. Where are participant data, analytics and public pages hosted?
6. Who may revoke or correct a credential, and what appeal process applies?
7. Will Basel describe the achievement as non-credit professional learning or as a formal microcredential?
8. Can a future Swiss wallet or EDC representation be generated without re-enrolling learners?

9. How will accessibility, multilingual content and name changes be handled?

13. Recommended next 90 days

Days 1–30: governance and test definition

- Nominate academic owner, credential registrar, privacy/security owner and Implement counterpart.
- Approve achievement title, outcomes, assessment, workload, evidence and explicit non-credit status.
- Choose Model C and draft issuer/endorser wording.
- Create a procurement test requiring OB3 export, independent verification, status, key-control documentation, GDPR/FADP terms and exit portability.

Days 31–60: interoperability build

- Compare Accredible, Open Badge Factory and one lightweight/open-source route.
- Generate test credentials from the prototype.
- Validate them with the 1EdTech verifier, VerifierPlus and two wallet/export pathways.
- Test key rotation, revoked credential, altered credential, expired credential and participant-name correction.

Days 61–90: live cohort preparation

- Enrol one Change Management cohort.
- Issue both Basel's current-style certificate and the OB3 credential for comparison.
- Instrument claim, sharing, export, verification and support metrics.
- Conduct learner and employer comprehension interviews.
- Produce a short go/no-go dossier covering interoperability, governance, privacy, administration, learner value and migration to Swiss/European infrastructure.

Final recommendation

Pilot now. Use the pilot to prove **portable verification and governance**, not to announce a universal credential ecosystem. Keep **University of Basel as the single signed issuer**, place **Implement Consulting Group visibly in delivery and achievement design**, and use a separate standards-based endorsement where feasible. Choose OB3 because it is lightweight enough for professional learning yet compatible with the broader VC direction; retain a PDF rendering; and preserve an eventual EDC/Swiss-wallet migration path.